Polyhedral spheres with congruent faces have an even number of faces

Fable 5.2 / prompting by Dan Horwitz

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Abstract. Let a polyhedral sphere in $\mathbb{R}^d$ be tiled edge to edge by congruent planar simple polygons, with at least three faces at every vertex. We prove that the number of faces is even. Neither convexity nor embeddedness is assumed, and the dimension is arbitrary. For convex polyhedra in $\mathbb{R}^3$ the statement is attributed to Grünbaum; the general question was raised by Hungerbühler, Hungerbühler and Pirron. Counting reduces the problem to parallelograms, which yield to a zone argument, and to kites. For kites the proof has three steps. The vertices are lifted to $\mathbb{R}^{d+1}$ by a height function that depends only on the combinatorics and on the shape of the kite, and the new coordinate is given the weight that makes two angles of every lifted face right angles. If this weight is positive, the lift is a stretch of the polyhedron and every face becomes inscribed in a circle; in general the lifted space is Lorentzian. The edge vectors round every lifted face then have Clifford holonomy -1. And on a sphere, central face holonomies multiply to 1. The last two steps prove more. A polyhedral sphere whose faces are convex quadrilaterals inscribed in circles has an even number of faces, and one whose faces are convex polygons with an even number of sides and equal sums of alternate angles has an even number of vertices. The mechanism is that of recent incidence theorems of Fomin and Pylyavskyy and of Izosimov, with the sign kept. These statements fail on the torus.

Note on provenance. The proof of the kite case was found on 12 November 2026 by an AI system (Claude Fable 5.2, Anthropic), prompted by the person named above, after a four-day programme in October that had reduced the problem and assembled the data. **No human mathematician has yet verified it.** This write-up is meant to make that verification easy. Corrections and references to earlier work are welcome, through the repository that holds this file.

1. Introduction

1.1. Statements

A *cell decomposition* of the sphere is a finite connected graph drawn on S^2, loops and multiple edges allowed, whose complementary regions, the *faces*, are open discs. A *polyhedral sphere in* $\mathbb{R}^d$ is a cell decomposition of S^2 together with a map X from its vertices to $\mathbb{R}^d$ such that, for every face, the images of its corners, joined in cyclic order by segments, form a planar simple polygon: a closed curve in a plane that does not meet itself. The sides of these polygons are the images of the edges, so the faces meet edge to edge. Nothing else is required. The surface may cross itself, neighbouring faces may be coplanar or folded onto each other, different vertices may have the same image, and a corner of a face may have a straight angle. The definition has some immediate consequences: the corners of a face are different vertices with different images, a face has at least three sides, there are no loops, and every edge lies on two different faces. Two faces are *congruent* if an isometry carries the corners of one to the corners of the other, in cyclic order in either direction; a corner with a straight angle counts as a corner.

Theorem 1. *Let a polyhedral sphere in* $\mathbb{R}^d$ *have congruent faces. If every vertex has degree at least* 3*, or if the faces are quadrilaterals, then the number of faces is even.*

For convex polyhedra in $\mathbb{R}^3$ the statement is attributed to Grünbaum [G]; we have not seen his paper and rely on [HHP]. Hungerbühler, Hungerbühler and Pirron [HHP] observed that in genus

0 only quadrilateral faces could occur in an odd number, excluded seven faces, and wrote: "It is an open question if monohedra with an odd number of quadrilateral faces exist." Theorem 1 answers the question for spheres in the edge-to-edge setting above, in every dimension. For edge-to-edge tilings of the round sphere by congruent spherical polygons, with all vertices of degree at least 3, the parity statement was proved by Luk [L]. His setting is different: at each vertex of a spherical tiling the angles add up to 2π, which has no counterpart for flat faces.

Counting and Euler's formula leave only quadrilateral faces whose side lengths occur in pairs: parallelograms and *kites*, where we use the word for every simple quadrilateral with two pairs of equal adjacent sides, convex or not. Parallelograms are settled by a zone argument. The case of kites is the content of this paper. Its proof gives more.

Call a convex polygon with an even number of sides *balanced* if its interior angles at every second corner have the same sum as its interior angles at the other corners. Straight angles are allowed and count as interior angles π. A convex polygon with an even number of sides that is inscribed in a circle is balanced, and a convex quadrilateral is balanced exactly when it is inscribed in a circle.

Theorem 2. *A polyhedral sphere in $\mathbb{R}^d$ all of whose faces are balanced convex polygons has an even number of vertices. Equivalently, the number of its faces whose number of sides is divisible by 4 is even. In particular, a polyhedral sphere all of whose faces are convex quadrilaterals inscribed in circles has an even number of faces.*

Here the faces need not be congruent, and the circles are unrelated. Only the last sentence is needed for Theorem 1. Faces with $6, 10, 14, ...$ sides are not counted, so the theorem says nothing about a sphere of hexagons.

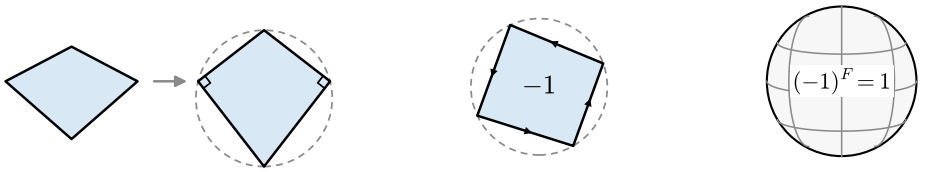
Theorem 2 is close to known results, and what is new in it is the conclusion about parity. Fomin and Pylyavskyy [FP, Corollary 11.6] consider a closed oriented surface tiled by quadrilaterals whose vertices are sent to distinct points of the plane, and show that if all tiles but one are inscribed in circles, then so is the last, because the product of the cross-ratios of the tiles is 1. Counting the signs in that product gives Theorem 2 for quadrilaterals in the plane. Izosimov [I, Theorem 7.3] proves the same statement for a tiling of the sphere with vertices in S^4, by means of a connection for which a face has trivial holonomy exactly when its cross-ratio is real; his Corollary 3.11 is the sphere lemma of Section 2 in the case that all face holonomies but one are trivial. Neither paper draws a conclusion about the number of faces inscribed in circles. The sign itself does occur in [FP, §10], where incidence theorems are proved from tilings that contain an even number of tiles with cross-ratio -1. That the cross-ratio of a quadrilateral in space is real exactly when it is inscribed in a circle, and negative exactly when the quadrilateral is embedded, is standard in discrete differential geometry [BP, Lemma 2]. Finally, for a convex polyhedron in $\mathbb{R}^3$ with all vertices on one sphere and a bipartite graph, the two colour classes of vertices have equal size, by an argument of Steinitz [S] (see [PZ, §1]), so the number of vertices is even. Theorem 2 needs neither convexity of the polyhedron nor a common sphere. It gives only the parity, and no more is true in this generality: the twelve squares of the 4-cube that avoid two opposite vertices form a polyhedral sphere in $\mathbb{R}^4$, with all its vertices on one 3-sphere, whose colour classes have sizes 8 and 6.

The kite case of Theorem 1 also has a purely algebraic form, valid in every signature and over every field of characteristic other than 2. It is Theorem 3 in Section 6.

1.2. The idea

Let the faces be congruent kites. The vertices carry a height function η, determined by the combinatorics and by the shape of the kite, such that the lifted points (X_v, η_v) in $\mathbb{R}^{d+1}$ again span planar faces. Measure the new coordinate with a weight τ. For one value of τ every lifted face is a kite with two right angles. Suppose first that this value is positive. Then the lift is a stretch of the polyhedron, and the corners of every lifted face lie on a circle (Figure 1). Now go round a face and compose, for each edge in turn, the reflection in the hyperplane orthogonal to that edge. For a convex quadrilateral inscribed in a circle the four reflections compose to one full turn, because opposite angles add up to π. A full turn is the identity on vectors and -1 on spinors. So in the Clifford algebra every face has holonomy -1, and on a sphere such central holonomies multiply to 1. Hence $(-1)^F = 1$.

The weight τ can also be zero or negative, and for darts it is always negative. The lifted space is then degenerate or Lorentzian, and there is no stretch and no circle, but the computation in the Clifford algebra is the same. On a torus the last step fails, and so do the theorems (Section 7).



1. stretch until the corners of every face lie on a circle

2. going round such a face gives the sign -1

3. on a sphere the signs multiply to $+1$

Figure 1: The three steps of the proof for kites, drawn for a positive weight τ.

2. The sphere lemma

Fix an orientation of S^2 and read the boundary of every face in the induced direction. Every edge is then traversed twice, once in each direction, by the faces on its two sides, which may be the same face.

Lemma 2.1 (sphere lemma). *Let a cell decomposition of S^2 be given, and let every traversal $x \to y$ of an edge carry an element $h(x \to y)$ of a group G, with $h(y \to x) = h(x \to y)^{-1}$ for the two directions of an edge. The* holonomy *of an edge path is the product of the elements along it, later ones on the left. If the holonomy round every face f is a central element z_f of G, then*

(i) the holonomy of every closed edge path lies in the subgroup generated by the z_f;
(ii) $\prod_f z_f = 1$.

A central holonomy does not depend on the corner at which the boundary path starts, because a cyclic shift of the path conjugates it.

Proof. Induct on the number of faces. *One face.* The graph has no cycle, since by the Jordan curve theorem a cycle would cut the sphere into two parts, each containing a face. So it is a tree. A closed walk in a tree, if not empty, contains an edge followed at once by its reverse, at a vertex farthest from the start. The two elements cancel, and removing them leaves a shorter closed walk. So every closed path has holonomy 1. This is (i), and it gives (ii) because the boundary of the face is such a walk.

More than one face. Some edge e has two different faces f and f' on its two sides. Otherwise each face together with the open edges on its boundary would be an open set, and these disjoint

sets would cover the sphere minus the vertices, which is connected. Let e run from x to y. The boundary of f is e followed by a path P from y back to x, and the boundary of f' is the reverse of e followed by a path Q from x back to y. Neither P nor Q uses e, because each of the two faces lies on one side of e only. So $z_f = h(P)h(e)$ and $z_{f'} = h(Q)h(e)^{-1}$.

Delete e (Figure 2). The graph stays connected, because P joins the ends of e. The faces f and f' merge into one region, which is an open disc, being two open discs joined along an open arc. Its boundary is P followed by Q, with holonomy

$$h(Q)h(P) = z_{f'}h(e) \cdot z_f h(e)^{-1} = z_f z_{f'},$$

which is central. The new decomposition has one face fewer and the same product of face holonomies, and its face holonomies lie in the subgroup generated by the old ones. By induction (ii) holds, and (i) holds for the closed paths that avoid e. A closed path that uses e turns into one that avoids e when every traversal of e is replaced by the reverse of P, and every traversal of the reverse of e by P. Since $h(e) = h(P)^{-1}z_f$, each replacement changes the holonomy by the central factor z_f or its inverse. So (i) holds for all closed paths. □

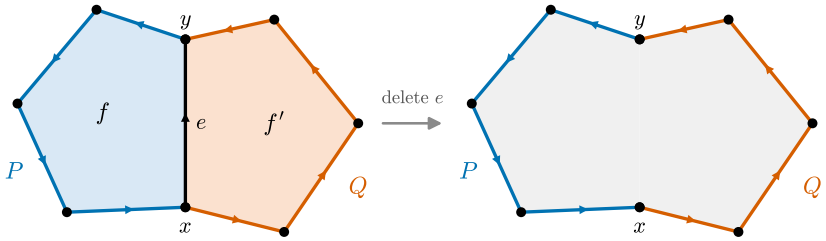


Figure 2: The induction step of the sphere lemma: deleting an edge between two faces.

Corollary 2.2 (two colours). *If every face of a cell decomposition of S^2 has an even number of sides, then the vertices can be coloured black and white so that every edge joins the two colours.*

Proof. Take $G = \{\pm 1\}$ and $h = -1$ on every traversal. Every face has holonomy 1, so by (i) every closed edge path has an even number of edges. Colour a vertex by the parity of the number of edges of a path that leads to it from a base vertex. □

Corollary 2.3 (heights). *Let κ be a real function on the traversals of the edges with $\kappa(y \to x) = -\kappa(x \to y)$, whose values add up to 0 round every face. Then there is a function η on the vertices with $\eta_y - \eta_x = \kappa(x \to y)$ for every traversal.*

Proof. Take $G = (\mathbb{R}, +)$. By (i) the values of κ add up to 0 along every closed path. So the sum of κ along a path from a base vertex to v depends only on v; call it η_v. □

3. Balanced faces

Let V be a vector space over a field of characteristic other than 2, with a symmetric bilinear form $\langle \cdot, \cdot \rangle$. Its Clifford algebra $\mathrm{Cl}(V)$ is the associative algebra with unit generated by V with the relations $vv = \langle v, v \rangle$. Then $uv + vu = 2\langle u, v \rangle$, orthogonal vectors anticommute, and a vector with $\langle v, v \rangle \neq 0$ is invertible with $v^{-1} = v/\langle v, v \rangle$. The algebra acts on the exterior algebra ΛV: a vector v acts as exterior multiplication by v plus contraction with $\langle v, \cdot \rangle$, and the square of this operator is $\langle v, v \rangle$. In this action -1 is minus the identity. So $-1 \neq 1$ in $\mathrm{Cl}(V)$, because the characteristic is not 2, whether or not the form is degenerate. Nothing else about Clifford algebras is used. In this section $V = \mathbb{R}^d$ with the Euclidean inner product.

Recall that a convex polygon with $2n$ sides is balanced if its interior angles at every second corner add up to half of the sum of all its interior angles, that is, to $(n-1)\pi$. A convex polygon with $2n$ sides inscribed in a circle is balanced. Its interior angle at a corner is π minus half the sum of the two arcs between that corner and its neighbours, and each of the $2n$ arcs between successive corners ends at exactly one of the corners $z_2, z_4, ..., z_{2n}$. So the angles at these n corners add up to $n\pi - \pi$.

Lemma 3.1. *Let $z_1, ..., z_{2n} \in \mathbb{R}^d$ be the successive corners of a balanced convex polygon, and put $g_i = z_{i+1} - z_i$, indices modulo $2n$. Then in $\mathrm{Cl}(\mathbb{R}^d)$*

$$g_{2n} g_{2n-1} \cdots g_2 g_1 = -|g_1|\,|g_2| \cdots |g_{2n}|.$$

Proof. Let e_1, e_2 be an orthonormal basis of the plane of directions of the polygon, chosen so that the corners run counterclockwise, and put $I = e_1 e_2$. Then $I^2 = -1$, and for vectors u, v of this plane $uv = |u||v|(\cos\theta + I \sin\theta)$, where θ is the angle from u to v. These products form a copy of the complex numbers; in particular they commute. The angle from g_{2k-1} to g_{2k} is the exterior angle $\pi - \beta_{2k}$ at z_{2k}, where β_i is the interior angle at z_i. So $g_{2k} g_{2k-1} = |g_{2k-1}|\,|g_{2k}|\, e^{-I(\pi - \beta_{2k})}$ for $k = 1, ..., n$. The product of these n elements is the left side of the lemma. It equals $|g_1| \cdots |g_{2n}|$ times $e^{-I(n\pi - \beta)}$, where $\beta = \beta_2 + \beta_4 + \cdots + \beta_{2n}$. The polygon is balanced, so $\beta = (n-1)\pi$, and the factor is $e^{-I\pi} = -1$. □

Proof of Theorem 2. All faces have an even number of sides. By Corollary 2.2 colour the vertices. For an edge with black end x and white end y let $g = X_y - X_x$, a nonzero vector, and put $h(x \to y) = g$ and $h(y \to x) = g^{-1}$. Read a face f with $2n$ sides, $n = n_f$, from a black corner, let $z_1, ..., z_{2n}$ be the images of its successive corners, and let g_i be as in Lemma 3.1. The traversals contribute g_1, then $(-g_2)^{-1} = -g_2/|g_2|^2$, then g_3, then $(-g_4)^{-1}$, and so on. So the holonomy round the face is

$$(-1)^n \frac{g_{2n} \cdots g_2 g_1}{|g_2|^2\,|g_4|^2 \cdots |g_{2n}|^2} = (-1)^{n+1} \rho_f, \quad \rho_f = \frac{|g_1|\,|g_3| \cdots |g_{2n-1}|}{|g_2|\,|g_4| \cdots |g_{2n}|} > 0,$$

a nonzero real number, which is central. By the sphere lemma the product of these numbers over all faces is 1. So the sum of the numbers $n_f + 1$ is even. The n_f add up to the number E of edges, since a face with $2n_f$ sides lies on $2n_f$ edges and every edge lies on two faces. So $E + F$ is even, and by Euler's formula $V = 2 + E - F$ is even. For the second form of the statement, $n_f + 1$ is odd exactly when the number $2n_f$ of sides of f is divisible by 4. If all faces are quadrilaterals, that is every face, so F is even. □

Remarks. (1) In the plane the argument for quadrilaterals is three lines. Take the vertices as complex numbers and put $c_f = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_2 - z_3)(z_4 - z_1)}$ for a face read from a black corner, in the direction fixed in Section 2. Every edge occurs in two faces, once in a numerator and once, reversed, in a denominator, so the product of the c_f is $(-1)^{2F} = 1$, as there are $2F$ edges. A convex quadrilateral inscribed in a circle has $c_f < 0$. That the product is 1 is the proof of [FP, Corollary 11.6]. Complex numbers commute, so no sphere lemma is needed: in the plane the statement for quadrilaterals holds on every closed oriented surface whose vertices can be two-coloured, as is noted for the incidence theorem in [I, §7]. The Clifford algebra carries the identity to higher dimensions, where the cross-ratios no longer commute and the sphere lemma is needed.

(2) For a crossed quadrilateral inscribed in a circle the angles at z_2 and z_4 are equal, since both stand on the chord $z_1 z_3$ from the same side, and the turns there have opposite senses.

The same computation then gives $+|g_1|\,|g_2|\,|g_3|\,|g_4|$. So if crossed faces are admitted among the quadrilaterals of Theorem 2, the number of convex ones is even.

(3) The step from the parity of $E+F$ to the faces with a number of sides divisible by 4 is Lemma 2 of Kuperberg [Ku], where it serves to show that a connected plane bipartite graph with an even number of vertices has a flat Kasteleyn weighting. The sign $(-1)^{n+1}$ of a face with $2n$ sides is the sign in that theory.

4. Kites: the stretch

4.1. Kites

Let $ASTU$ be a planar simple quadrilateral with $|AS| = |AU| = a$ and $|TS| = |TU| = b$, where $a \neq b$. Then $A \neq T$, both points lie on the perpendicular bisector of SU, and the midpoint of SU is $(1-\frac{r}{2})A + (\frac{r}{2})T$ for one real number r, the *shape* of the quadrilateral. So

$$S + U = (2-r)A + rT. \tag{1}$$

The quadrilateral is a convex kite for $0 < r < 2$ and a dart for $r < 0$ or $r > 2$. For $r = 0$ the angle at A is straight and for $r = 2$ the angle at T: the polygon is then an isosceles triangle with the midpoint of its base as a fourth corner (Figure 3). The angles at S and U are never straight. We write $\sigma = |SU|$ and $\alpha = |AT|$, both positive.

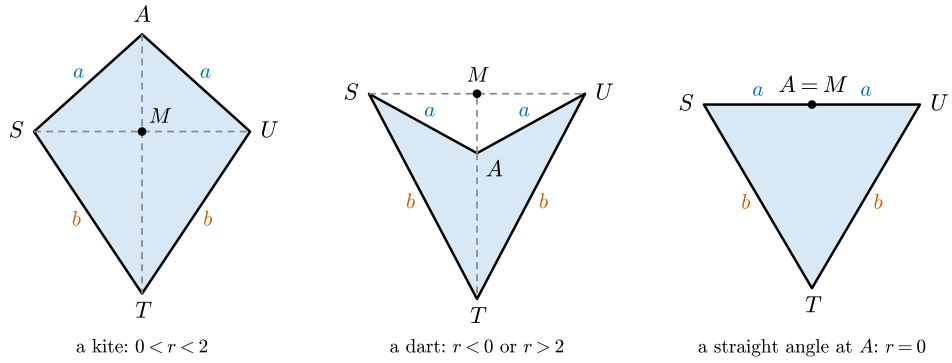


Figure 3: The three kinds of face in the kite case. M is the midpoint of SU.

Consider a polyhedral sphere whose faces are congruent to $ASTU$. Label the corners of every face accordingly; the labels are determined up to exchanging S and U. Call an edge an a-edge or a b-edge according to its length. In every face AS and AU are a-edges and ST and TU are b-edges. By Corollary 2.2 colour the vertices black and white. In every face A and T have one colour and S and U the other; put $\delta_f = +1$ if A is black and $\delta_f = -1$ if A is white.

Proposition 4.1. *A polyhedral sphere in $\mathbb{R}^d$ whose faces are congruent kites of a shape $r \notin \{0, 2\}$ has an even number of faces.*

4.2. The height function

For a traversal $x \to y$ of an edge with x black put $\kappa(x \to y) = r$ on an a-edge and $\kappa(x \to y) = -(2-r)$ on a b-edge, and put $\kappa(y \to x) = -\kappa(x \to y)$. Round a face,

$$\kappa(A \to S) = \delta r, \quad \kappa(S \to T) = \delta(2-r), \quad \kappa(T \to U) = -\delta(2-r), \quad \kappa(U \to A) = -\delta r,$$

with $\delta = \delta_f$, and these add up to 0. By Corollary 2.3 there is a function η on the vertices with these increments. In every face

$$\eta_S = \eta_U = \eta_A + \delta r, \quad \eta_T = \eta_A + 2\delta, \tag{2}$$

so η satisfies the relation (1): $\eta_S + \eta_U = (2-r)\eta_A + r\eta_T$.

4.3. The lift

Let

$$\tau = \frac{\langle X_A - X_S, X_T - X_S\rangle}{r(2-r)} = \frac{1}{4}\left(\frac{\sigma^2}{r(2-r)} - \alpha^2\right),$$

the same number for every face. On $\mathbb{R}^{d+1} = \mathbb{R}^d \oplus \mathbb{R}$ take the symmetric bilinear form $\langle (x,s),(y,t)\rangle_\tau = \langle x, y\rangle + \tau st$, and lift every vertex to $\hat{X}_v = (X_v, \eta_v)$.

Lemma 4.2. *In a face put $p = \hat{X}_S - \hat{X}_A$, $q = \hat{X}_T - \hat{X}_S$, $p' = \hat{X}_U - \hat{X}_T$ and $q' = \hat{X}_A - \hat{X}_U$. Then*

(a) $p' = -(2-r)p - (1-r)q$ and $q' = (1-r)p - rq$;
(b) $\langle p, q\rangle_\tau = 0$;
(c) $\langle p, p\rangle_\tau = \langle q', q'\rangle_\tau = N_a$ and $\langle q, q\rangle_\tau = \langle p', p'\rangle_\tau = N_b$, where $N_a = \frac{\sigma^2}{2(2-r)}$ and $N_b = \frac{\sigma^2}{2r}$.

Proof. (a) Both X and η satisfy (1), so $\hat{X}_U = (2-r)\hat{X}_A + r\hat{X}_T - \hat{X}_S$. Substituting this in p' and q' gives the two formulas.

(b) By (2), $\langle p, q\rangle_\tau = \langle X_S - X_A, X_T - X_S\rangle + \tau \cdot \delta r \cdot \delta(2-r) = -\tau r(2-r) + \tau r(2-r) = 0$.

(c) The diagonals of the kite are orthogonal, and the midpoint of SU lies at the signed distances $(r/2)\alpha$ from A and $(1-r/2)\alpha$ from T along the axis. So $|X_S - X_A|^2 = \sigma^2/4 + r^2\alpha^2/4$ and

$$\langle p, p\rangle_\tau = \frac{\sigma^2}{4} + \frac{r^2\alpha^2}{4} + \tau r^2 = \frac{\sigma^2}{4} + \frac{r\sigma^2}{4(2-r)} = \frac{\sigma^2}{2(2-r)}.$$

In the same way $\langle q, q\rangle_\tau = \sigma^2/4 + (2-r)^2\alpha^2/4 + \tau(2-r)^2 = \sigma^2/(2r)$. The edges UA and TU have the same lengths as AS and ST and, up to sign, the same height differences, which gives the values for q' and p'. □

So every lifted face is a planar quadrilateral, by (a), with a right angle at S, by (b), and in the same way at U. Note that N_a and N_b are not zero, that $N_b < 0$ for a dart with $r < 0$, and that $N_a < 0$ for a dart with $r > 2$.

Suppose that $\tau > 0$; then $0 < r < 2$. After the last coordinate is rescaled by $\sqrt{\tau}$ the form is the standard inner product of $\mathbb{R}^{d+1}$, so the lift is obtained by stretching along the new axis (Figure 4). In a lifted face the points $\hat{X}_S$ and $\hat{X}_U$ lie on the circle with diameter $\hat{X}_A\hat{X}_T$, and on different sides of that diameter: by (1) the midpoint of $\hat{X}_S\hat{X}_U$ lies on the line through $\hat{X}_A$ and $\hat{X}_T$, and $\hat{X}_S$ does not, because p and q are orthogonal and not zero. So every lifted face is a convex quadrilateral inscribed in a circle, the lift is a polyhedral sphere in $\mathbb{R}^{d+1}$, and Proposition 4.1 follows from Theorem 2. If $\tau = 0$ the given faces have right angles at S and U already, and Theorem 2 applies to them. If $\tau < 0$, which is always so for darts, the form is Lorentzian. The following computation covers all cases.

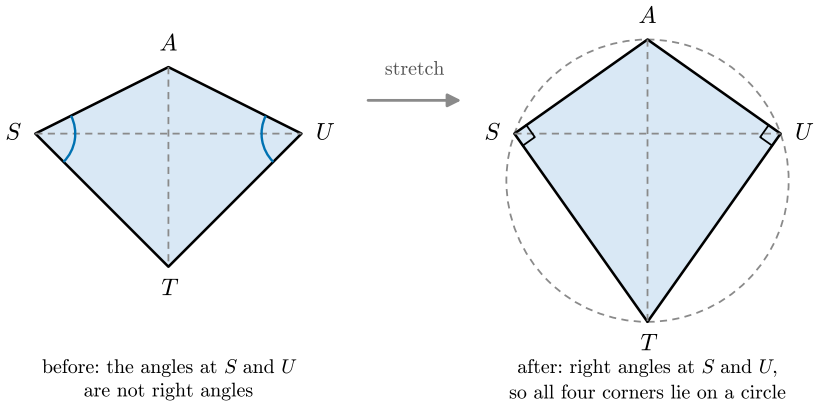


Figure 4: The effect of the lift on one face when $\tau > 0$: the axis AT is lengthened, the side diagonal SU and the shape r are kept.

Lemma 4.3. *In the Clifford algebra of $\mathbb{R}^{d+1}$ with the form $\langle \cdot, \cdot \rangle_\tau$, and with the notation of Lemma 4.2, $q'p'qp = -N_a N_b$.*

Proof.

By Lemma 4.2, $p^2 = N_a$, $q^2 = N_b$ and $qp = -pq$. So

$$q'p' = ((1-r)p - rq)(-(2-r)p - (1-r)q)$$
$$= [-(1-r)(2-r)N_a + r(1-r)N_b] - [(1-r)^2 + r(2-r)]pq = -pq,$$

because the first bracket is $-(1-r)\sigma^2/2 + (1-r)\sigma^2/2 = 0$ and the second is 1. Then $q'p'qp = -pqqp = -N_a N_b$. □

Proof of Proposition 4.1. For an edge with black end x and white end y let $g = \hat{X}_y - \hat{X}_x$. By Lemma 4.2 (c), $\langle g, g \rangle_\tau$ is N_a or N_b, so g is invertible in the Clifford algebra; put $h(x \to y) = g$ and $h(y \to x) = g^{-1}$. Take the path $A \to S \to T \to U \to A$ round a face. If $\delta_f = +1$ the traversals contribute p, $(-q)^{-1} = -q/N_b$, p' and $(-q')^{-1} = -q'/N_a$, with product $q'p'qp/(N_a N_b) = -1$ by Lemma 4.3. If $\delta_f = -1$ they contribute $-p/N_a$, q, $-p'/N_b$ and q', with the same product. The reversed path has the inverse holonomy, also -1. By the sphere lemma $(-1)^F = 1$, and since $-1 \neq 1$ in the Clifford algebra, F is even. □

5. Proof of Theorem 1

Two shapes remain after Proposition 4.1: the kites with a straight angle, and the parallelograms.

Proposition 5.1. *A polyhedral sphere in $\mathbb{R}^d$ whose faces are congruent kites of shape $r = 0$ or $r = 2$ has an even number of faces.*

Proof. Let $r = 0$, so that X_A is the midpoint of X_S and X_U in every face. For an a-edge e with black end x and white end y put $g_e = X_y - X_x$, a nonzero vector. The two a-sides of a face are AS and AU, and A is the black end of both or the white end of both. So $g_{AS} = -g_{AU}$. Every a-edge lies on two faces and every face has two a-sides, so the faces fall into closed chains $f_1, e_1, f_2, e_2, \ldots, f_n, e_n$ in which the a-sides of f_i are e_{i-1} and e_i, indices modulo n. Going once round such a chain, $g_{e_n} = (-1)^n g_{e_n}$. So every chain has an even number of faces, and F is even. For $r = 2$ repeat the argument with T in place of A and the b-edges in place of the a-edges. □

In a polyhedron the half-bases met by one chain lie on a line, and their vectors from black to white alternate. The proposition uses the sphere only for the two colours.

Proof of Theorem 1. Suppose that F is odd.

Counting. If the polygon has m sides of length ℓ, then mF is twice the number of edges of length ℓ. So every such multiplicity m is even, and the number n of sides is even.

Euler. Let V and E be the numbers of vertices and edges. If every vertex has degree at least 3, then $3V \le 2E = nF$ and $2 = V - E + F \le F(1 - n/6)$, so $n \le 5$ and hence $n = 4$. This is the only use of the hypothesis on the degrees.

Shapes. A quadrilateral in which every side length occurs an even number of times has the cyclic pattern (a, b, a, b), where $a = b$ is allowed, or (a, a, b, b) with $a \neq b$. In the first case let the corners be P_1, P_2, P_3, P_4. The triangles $P_1P_2P_3$ and $P_3P_4P_1$ are congruent, so P_4 is $P_1 + P_3 - P_2$ or the mirror image of this point in the line P_1P_3. If P_2 is not on the line P_1P_3, then with the second choice two sides cross or P_4 coincides with P_2; if P_2 is on that line, the two choices are the same point. So a simple quadrilateral of this pattern is a parallelogram. It has no straight angle: opposite sides are parallel, so if three successive corners lay on one line, all four would. The second pattern is that of Section 4.

Parallelograms. A *zone* is a closed strip of faces in which each face is joined to the next through opposite sides. Every edge is crossed by exactly one zone, and every face is crossed twice, once through each pair of opposite sides. Opposite sides of a parallelogram are equal as vectors, so all edges crossed by one zone are equal up to sign. If both crossings of a face belonged to one zone, all four sides of that face would be parallel and its corners would lie on one line. So no zone crosses itself: the curve that joins the midpoints of the edges crossed by a zone is a simple closed curve on S^2. By the Jordan curve theorem two such curves cross an even number of times. Every face is a crossing of two different zones, and they cross once there. So F is even, a contradiction.

Kites. In the pattern (a, a, b, b) with $a \neq b$ the corner A is the one between the two sides of length a, and the setting of Section 4 applies, with one shape r for all faces. Proposition 4.1, or Proposition 5.1 if r is 0 or 2, gives that F is even, a contradiction. □

6. The kite case for quadratic forms

The proof of Proposition 4.1 used very little of the geometry: relation (1) for the coordinates, and equal side diagonals. This can be made precise, and the statement that results needs neither positivity nor the real numbers.

A *kite labelling* is a cell decomposition of S^2 into quadrilateral faces whose corners carry the labels A, S, T, U in cyclic order, such that every edge has one type, the same on its two sides, also when both sides lie in one face. Here AS and AU have type a, and ST and TU have type b. Multiple edges and vertices of degree 2 are allowed. Let $\Bbbk$ be a field of characteristic other than 2 and $r \in \Bbbk$. Let $K(r)$ be the space of functions x from the vertices to $\Bbbk$ with $x_S + x_U = (2 - r)x_A + rx_T$ on every face, let $W = K(r)^*$ be its dual space, let $\varepsilon_v \in W$ be the evaluation at the vertex v, and put $s_f = \varepsilon_S - \varepsilon_U$.

Theorem 3. *Let $\Bbbk$ be a field of characteristic other than 2 and $r \in \Bbbk$. If, for a kite labelling of the sphere, there is a symmetric bilinear form Γ on $W = K(r)^*$ with $\Gamma(s_f, s_f) = 1$ for every face f, then the number of faces is even.*

For a polyhedral sphere of congruent kites the coordinate functions $x^1, ..., x^d$ of X lie in $K(r)$, and $\Gamma(\varphi, \psi) = \sigma^{-2} \sum_j \varphi(x^j)\psi(x^j)$ is such a form. So Theorem 3 contains Propositions 4.1 and 5.1.

Proof. For $r \in \{0,2\}$ the proof of Proposition 5.1 applies with $g_e = \varepsilon_y - \varepsilon_x \in W$. A chain may now consist of one face whose two a-sides are one edge. An odd chain gives $2g_e = 0$ and hence $s_f = \pm 2g_e = 0$ for its faces, against $\Gamma(s_f, s_f) = 1$.

Let $r \notin \{0,2\}$. Colour the vertices and define δ_f and $\kappa(x \to y)$ as in Section 4. In this proof write $N_a = 1/(2(2-r))$ and $N_b = 1/(2r)$, the values of Lemma 4.2 for $\sigma = 1$, and N_e for the one that belongs to the type of the edge e. Let $W' = W \oplus \Bbbk\nu \oplus \Bbbk\mu$, and extend Γ to a symmetric form Γ' on W' in which ν and μ are orthogonal to W, with $\Gamma'(\nu,\nu) = \Gamma'(\mu,\mu) = 0$ and $\Gamma'(\nu,\mu) = 1$. For a traversal $x \to y$ of an edge e, with $w = \varepsilon_y - \varepsilon_x$, put

$$v(x \to y) = w + \kappa(x \to y)\nu + c\mu, \quad c = \frac{N_e - \Gamma(w,w)}{2\kappa(x \to y)}.$$

Then $v(y \to x) = -v(x \to y)$, and $\Gamma'(v,v) = \Gamma(w,w) + 2\kappa c = N_e$. Moreover $v(x \to y)$ is the only element of the form $w + \kappa(x \to y)\nu + c'\mu$ with this norm.

In a face put $p = v(A \to S)$ and $q = v(S \to T)$. The element $(2-r)p - rq$ has W-part $2\varepsilon_S - (2-r)\varepsilon_A - r\varepsilon_T = s_f$ and ν-coefficient $(2-r)\delta r - r\delta(2-r) = 0$. So its norm is $\Gamma(s_f, s_f) = 1$. Expanding the same norm gives $(2-r)^2 N_a + r^2 N_b - 2r(2-r)\Gamma'(p,q) = 1 - 2r(2-r)\Gamma'(p,q)$. Hence $\Gamma'(p,q) = 0$. Next, $-(2-r)p - (1-r)q$ has W-part $\varepsilon_U - \varepsilon_T$, ν-coefficient $-\delta(2-r) = \kappa(T \to U)$ and norm $(2-r)^2 N_a + (1-r)^2 N_b = N_b$, so by uniqueness it is $v(T \to U)$. In the same way $(1-r)p - rq = v(U \to A)$. These are the statements of Lemma 4.2 for the elements v, with $\sigma = 1$. Lemma 4.3 and the proof of Proposition 4.1 now apply word for word in the Clifford algebra of W' with the form Γ', with $h(x \to y) = v(x \to y)$ for x black and $h(y \to x) = v(x \to y)^{-1}$: every face has holonomy -1, and the sphere lemma gives $(-1)^F = 1$. □

Here the height function is never constructed: only its increments κ enter. In the geometric case $c = \frac{\tau}{2\sigma^2}\kappa$, and $\nu + \frac{\tau}{2\sigma^2}\mu$ spans the one extra coordinate of Section 4. Every division in the proof is by 2, r, $2-r$ or a product of these, and no square root is taken.

7. Examples, limits and questions

The torus. Theorems 1, 2 and 3 fail on the torus, in two ways. Let Δ be an equilateral triangle. The nine squares of $\partial\Delta \times \partial\Delta \subset \mathbb{R}^4$ form an embedded torus of congruent squares; its graph contains triangles and cannot be two-coloured, so the edge elements are not defined. In $(\partial\Delta)^3 \subset \mathbb{R}^6$, with vertices indexed by $v \in (\mathbb{Z}/3)^3$ and with u_1, u_2, u_3 the standard basis, the 27 squares with corners v, $v + u_i$, $v + u_i + u_j$, $v + u_j$, where $i < j$ and $v_1 + v_2 + v_3 \equiv 0$, form a torus of congruent squares whose vertices can be two-coloured. Every face is inscribed in a circle and has holonomy -1, and $F = 27$ is odd. Only the sphere lemma fails: the product of the 27 face holonomies is -1. It is the commutator of the holonomies of two generating cycles of the torus, which anticommute here.

The hypotheses. Planarity of the faces is needed, at least when vertices of degree 2 are allowed. Take an equilateral triangle $Y_1Y_2Y_3$ and two different points Z, Z' on its axis, at different distances from its plane. The three skew quadrilaterals $ZY_iZ'Y_j$ are congruent, have the side lengths of a kite, and form a sphere. We have not looked for such an example in which every vertex has degree at least 3. In Theorem 3 the condition that every edge has one type is needed. The three faces $(0,3,1,4)$, $(0,4,2,3)$, $(1,3,2,4)$ on five vertices, labelled in this order, have a form Γ with $\Gamma(s_f, s_f) = 1$ for every r, and two of their edges have two types.

Not covered. (1) Polyhedra whose faces do not meet edge to edge, so that a corner of one face lies inside a side of another. Declaring every such point a corner of the face it lies on, with a

straight angle, makes the faces meet edge to edge. But Theorem 1 then applies only if the faces are still congruent with their corners, that is, if every face receives the same extra corners. In general they do not, and already the counting step fails. (2) Faces with six or more sides when vertices of degree 2 are allowed. At such a vertex only two faces meet: it is either a point inside a common side of both, or the two faces lie in one plane. Euler's formula then no longer limits the number of sides.

Questions. (1) Is there a torus in $\mathbb{R}^3$, embedded or not, with an odd number of congruent faces? The examples above live in dimensions 4 and 6. (2) Theorem 1 does not cover congruent faces with six or more sides, which can occur only together with vertices of degree 2. Is the number of faces even there as well? Theorem 2 does not help, since faces with $6, 10, 14, ...$ sides have a positive holonomy. (3) We have not found the parity statement of Theorem 2 in print, in any dimension; the nearest results are those quoted in the introduction. Is it known to the experts? (4) Does Theorem 1 hold for congruent faces that need not meet edge to edge? We know no example with an odd number of faces.

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